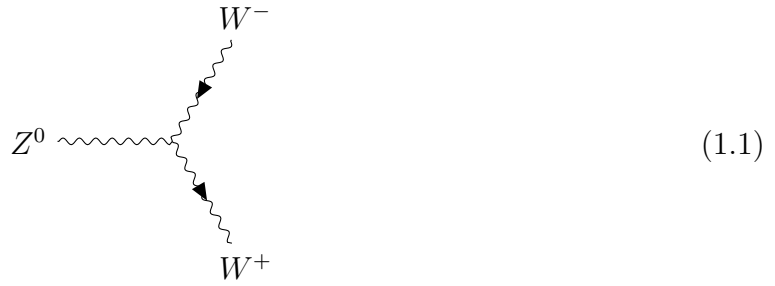


Problem 1 $Z \rightarrow W^+W^-$

If the W bosons had much less mass such that $Z \rightarrow W^+W^-$ were kinematically possible, compute what the particle width of $Z \rightarrow W^+W^-$ would be. You can assume the W is massless in this calculation. In this case, what would the ratio of $\Gamma(Z \rightarrow W^+W^-)/\Gamma(Z \rightarrow e^+e^-)$ be?



Problem 2 *Coupling Crossing*

Take a look at problem 10.6 in the textbook. For the Standard Model and the Minimal Supersymmetric Standard Model compute the energy scale where the $SU(2)$ coupling crosses the $U(1)$ coupling, and where the $SU(2)$ coupling crosses the $SU(3)$ coupling. Thus, you should compute four energy scales in this problem. You should compute these numbers analytically first and then plug in the numbers given in problem 10.6 to give numerical values in terms of GeV.

$$\mu \frac{dg_i}{d\mu} = \frac{B_i}{16\pi^2} g_i^3 \rightarrow \int_{g_i(M_Z)}^{g_i} \frac{dg'_i}{g_i'^3} = \frac{B_i}{16\pi^2} \int_{M_Z}^{\mu} \frac{d\mu'}{\mu'} \rightarrow \left. \frac{-1}{2g_i'^2} \right|_{g_i(M_Z)}^g = \frac{B_i}{16\pi^2} \ln \mu' \Big|_{M_Z}^{\mu} \quad (2.1)$$

$$\frac{-1}{2g_i(\mu)^2} + \frac{1}{2g_i(M_Z)^2} = \frac{B_i}{16\pi^2} \ln \left(\frac{\mu}{M_Z} \right) \rightarrow \boxed{\frac{g_i(\mu)^2}{4\pi} = \left[\frac{4\pi}{g_i(M_Z)^2} - \frac{B_i}{2\pi} \ln \left(\frac{\mu}{M_Z} \right) \right]^{-1}} \quad (2.2)$$

$U(1)_Y, SU(2)_L$ coupling crossing, $g_1(\mu) \propto U(1)_Y$ coupling and $g_2(\mu) = SU(2)_L$ coupling. Solving analytically for the intersection between any two couplings $g_i(\mu)$ and $g_j(\mu)$ where $i \neq j$:

$$\frac{g_i(\mu_c)^2}{4\pi} = \left[\frac{4\pi}{g_i(M_Z)^2} - \frac{B_i}{2\pi} \ln \left(\frac{\mu_c}{M_Z} \right) \right]^{-1} = \frac{g_j(\mu_c)^2}{4\pi} = \left[\frac{4\pi}{g_j(M_Z)^2} - \frac{B_j}{2\pi} \ln \left(\frac{\mu_c}{M_Z} \right) \right]^{-1} \quad (2.3)$$

$$\left[\frac{4\pi}{g_i(M_Z)^2} - \frac{B_i}{2\pi} \ln \left(\frac{\mu_c}{M_Z} \right) \right]^{-1} = \left[\frac{4\pi}{g_j(M_Z)^2} - \frac{B_j}{2\pi} \ln \left(\frac{\mu_c}{M_Z} \right) \right]^{-1} \quad (2.4)$$

$$\frac{4\pi}{g_i(M_Z)^2} - \frac{B_i}{2\pi} \ln \left(\frac{\mu_c}{M_Z} \right) = \frac{4\pi}{g_j(M_Z)^2} - \frac{B_j}{2\pi} \ln \left(\frac{\mu_c}{M_Z} \right) \quad (2.5)$$

$$\frac{4\pi}{g_i(M_Z)^2} - \frac{4\pi}{g_j(M_Z)^2} = -\frac{B_j}{2\pi} \ln \left(\frac{\mu_c}{M_Z} \right) + \frac{B_i}{2\pi} \ln \left(\frac{\mu_c}{M_Z} \right) \quad (2.6)$$

$$\frac{8\pi^2}{B_i - B_j} \left(\frac{1}{g_i(M_Z)^2} - \frac{1}{g_j(M_Z)^2} \right) = \ln \left(\frac{\mu_c}{M_Z} \right) \quad (2.7)$$

$$\boxed{\mu_c = M_Z \exp \left\{ \frac{8\pi^2}{B_i - B_j} \left(\frac{1}{g_i(M_Z)^2} - \frac{1}{g_j(M_Z)^2} \right) \right\}} \quad (2.8)$$

For the crossing between $SU(2)_L, U(1)_Y$:

$$\mu_c = M_Z \exp \left\{ \frac{8\pi^2}{B_2 - B_1} \left(\frac{1}{g_2(M_Z)^2} - \frac{1}{g_1(M_Z)^2} \right) \right\} \quad (2.9)$$

For finding the crossing between $SU(2)_L, SU(3)_c$, I just need to rewrite 2.8 in terms of $\alpha_3(\mu)$:

$$\mu_c = M_Z \exp \left\{ \frac{2\pi}{B_3 - B_2} \left(\frac{1}{\alpha_3(M_Z)} - \frac{4\pi}{g_2(M_Z)^2} \right) \right\} \quad (2.10)$$

For the Standard model:

$$\mathbf{i=1} \quad g_1(M_Z) = 0.461, \quad B_1 = 41/10$$

$$\mathbf{i=2} \quad g_2(M_Z) = 0.652, \quad B_2 = -19/6$$

$$\mathbf{i=3} \quad \alpha_3(M_Z) = 0.1185, \quad B_3 = -7$$

$$U(1)_Y \text{ coupling crossing } SU(2)_L \text{ coupling in SM at } \mu_c \approx 1.158 \times 10^{13} \text{ GeV} \quad (2.11)$$

$$SU(2)_L \text{ coupling crossing } SU(3)_c \text{ coupling in SM at } \mu_c \approx 9.897 \times 10^{16} \text{ GeV} \quad (2.12)$$

For the Minimally Supersymmetric Standard Model (MSSM):

$$\mathbf{i=1} \quad g_1(M_Z) = 0.461, \quad B_1 = 33/5$$

$$\mathbf{i=2} \quad g_2(M_Z) = 0.652, \quad B_2 = 1$$

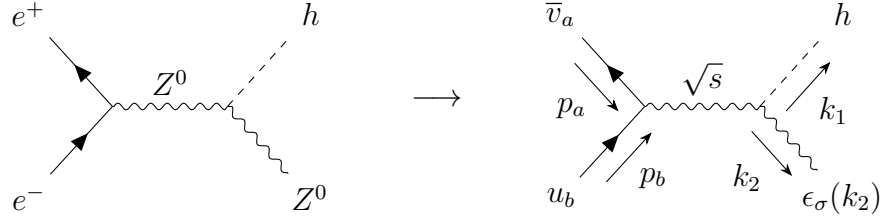
$$\mathbf{i=3} \quad \alpha_3(M_Z) = 0.1185, \quad B_3 = -3$$

$$U(1)_Y \text{ coupling crossing } SU(2)_L \text{ coupling in MSSM at } \mu_c \approx 2.336 \times 10^{16} \text{ GeV} \quad (2.13)$$

$$SU(2)_L \text{ coupling crossing } SU(3)_c \text{ coupling in MSSM at } \mu_c \approx 2.339 \times 10^{16} \text{ GeV} \quad (2.14)$$

Problem 3 $e^+e^- \rightarrow hZ$

Compute the cross-section of $e^+e^- \rightarrow hZ$ at energies $\sqrt{s} > m_h + M_Z$. Compute analytically first, and then find the numerical value of the cross-section in fb for $\sqrt{s} = 250\text{GeV}$, 500 GeV , 1TeV . At each of these energies, determine how many $b\bar{b}e^+e^-$ events occur that originate from this process followed by $h \rightarrow b\bar{b}$ and $Z \rightarrow e^+e^-$ decays. Use the following values for branching fractions: $B(h \rightarrow b\bar{b}) = 0.584$ and $B(Z \rightarrow e^+e^-) = 0.034$. Keep the masses of h and Z for this problem.



Matrix Element:

$$\mathcal{M} = \bar{v}_a(-ie\gamma^\nu)u_b \frac{i}{s - M_Z^2} \left[g_{\mu\nu} + \frac{p_\mu p_\nu}{M_Z^2} \right] i g^{\nu\sigma} \frac{(g^2 + g'^2)v}{2} \epsilon_\sigma(k_2) \quad (3.1)$$

$$= \frac{2M_Z^2}{v} \frac{ie}{s - M_Z^2} \bar{v}_a \gamma^\mu u_b \left[g_{\mu\nu} + \frac{p_\mu p_\nu}{M_Z^2} \right] g^{\nu\sigma} \epsilon_\sigma(k_2) \quad (3.2)$$

$$= \frac{2ieM_Z^2}{v(s - M_Z^2)} \epsilon^\nu(k_2) \left[\bar{v}_a \gamma_\nu u_b + \bar{v}_a \not{p}_Z u_b \frac{p_\nu}{M_Z^2} \right] \quad (3.3)$$

Problem 4 *Exotic Higgs Decay*

Let us assume that the Higgs boson can decay into two exotic massless X -bosons via a $h - X_\alpha - X_\beta$ Feynman rule of $i(g_{\alpha\beta}k_1 \cdot k_2 - k_{1\alpha}k_{2\beta})/\Lambda$, where Λ is a mass parameter, and k_1 (k_2) is the outgoing momentum of the X_α (X_β) boson. Compute the partial width of $h \rightarrow XX$. Determine the value of Λ such that the partial width of the Higgs boson into X -bosons is the same as the partial width of the Higgs boson into muons. [Note: there are 5th-force issues with an exotic photon being exactly massless, but for this problem you should treat it as effectively massless.]

$$(4.1)$$

$$\mathcal{M} = \frac{i(g_{\alpha\beta}(k_1 \cdot k_2) - k_{1\alpha}k_{2\beta})}{\Lambda} \epsilon_1^\alpha \epsilon_2^\beta \quad (4.2)$$

$$\sum_{\text{spins}} |\mathcal{M}|^2 = \sum_{\text{spins}} \frac{(g_{\alpha\beta}(k_1 \cdot k_2) - k_{1\alpha}k_{2\beta})}{\Lambda} \epsilon_1^\alpha \epsilon_2^\beta \frac{(g_{\mu\nu}(k_1 \cdot k_2) - k_{1\mu}k_{2\nu})}{\Lambda} \epsilon_1^{\mu*} \epsilon_2^{\nu*} \quad (4.3)$$

$$= \frac{1}{\Lambda^2} (g_{\alpha\beta}(k_1 \cdot k_2) - k_{1\alpha}k_{2\beta})(g_{\mu\nu}(k_1 \cdot k_2) - k_{1\mu}k_{2\nu}) g^{\alpha\mu} g^{\beta\nu} \quad (4.4)$$

$$= \frac{1}{\Lambda^2} \left(g^{\alpha\mu} g^{\beta\nu} g_{\alpha\beta}(k_1 \cdot k_2) g_{\mu\nu}(k_1 \cdot k_2) - g^{\alpha\mu} g^{\beta\nu} g_{\alpha\beta}(k_1 \cdot k_2) k_{1\mu} k_{2\nu} - \dots \right. \quad (4.5)$$

$$\left. \dots - g^{\alpha\mu} g^{\beta\nu} k_{1\alpha} k_{2\beta} g_{\mu\nu}(k_1 \cdot k_2) + g^{\alpha\mu} g^{\beta\nu} k_{1\alpha} k_{2\beta} k_{1\mu} k_{2\nu} \right)$$

$$= \frac{1}{\Lambda^2} \left(4(k_1 \cdot k_2)^2 - (k_1 \cdot k_2)(k_1 \cdot k_2) - \dots \right. \quad (4.6)$$

$$\left. \dots - (k_1 \cdot k_2)(k_1 \cdot k_2) + k_{1\alpha} k_{2\beta} k_1^\alpha k_2^\beta \right)$$

$$= \frac{2(k_1 \cdot k_2)^2 + (k_1 \cdot k_1)(k_2 \cdot k_2)}{\Lambda^2} \quad (4.7)$$

The outgoing momenta are equal, and the exotic bosons are massless, so the energy should add up to the higgs rest mass:

$$k_1^\mu = \frac{1}{2}[m_h, 0, 0, m_h] \quad k_2^\mu = \frac{1}{2}[m_h, 0, 0, -m_h] \quad (4.8)$$

$$k_1 \cdot k_2 = \frac{1}{2}m_h^2 \quad k_1 \cdot k_1 = k_2 \cdot k_2 = 0 \quad (4.9)$$

Plugging this in to the reduced matrix element:

$$\sum_{\text{spins}} |\mathcal{M}|^2 = \frac{m_h^4}{2\Lambda^2} \quad (4.10)$$

Now calculating the differential decay width:

$$d\Gamma = \frac{1}{64\pi^2 m_h} |\mathcal{M}|^2 d\phi_1 d\cos(\theta_1) \quad (4.11)$$

$$= \frac{1}{64\pi^2 m_h} \frac{m_h^4}{2\Lambda^2} d\phi_1 d\cos(\theta_1) \quad (4.12)$$

$$= \frac{m_h^3}{128\pi^2 \Lambda^2} d\phi_1 d\cos(\theta_1) \quad (4.13)$$

Using Martin-Wells (6.25) for the angular integrals (and remembering that the particles are indistinguishable):

$$\Gamma(h \rightarrow XX) = \frac{m_h^2}{128\pi^2 \Lambda^2} 2\pi \quad (4.14)$$

$$\Gamma(h \rightarrow XX) = \frac{m_h^2}{64\pi \Lambda^2} \quad (4.15)$$

What we want is to find the Λ that makes the following equation true:

$$1 = \frac{BR(h \rightarrow XX)}{BR(h \rightarrow \mu^+ \mu^-)} \quad (4.16)$$

$$1 = \frac{m_h^2}{64\pi \Lambda^2} \frac{1}{BR(h \rightarrow \mu^+ \mu^-)} \quad (4.17)$$

$$\Lambda = \frac{1}{\sqrt{BR(h \rightarrow \mu^+ \mu^-)}} \sqrt{\frac{m_h^2}{64\pi}} \quad (4.18)$$

From PDG: A Higgs with $m_h = 125\text{GeV}$ has a branching ratio to muons

$$BR(h \rightarrow \mu^+ \mu^-) = \frac{\Gamma(h \rightarrow \mu^+ \mu^-)}{\Gamma} = 2.6 \times 10^{-4}.$$

This means that the mass parameter Λ that would make the two branching ratios equal is:

$$\Lambda = \sqrt{\frac{(125)^2}{64 * \pi * 2.6 \times 10^{-4}}} \quad (4.19)$$

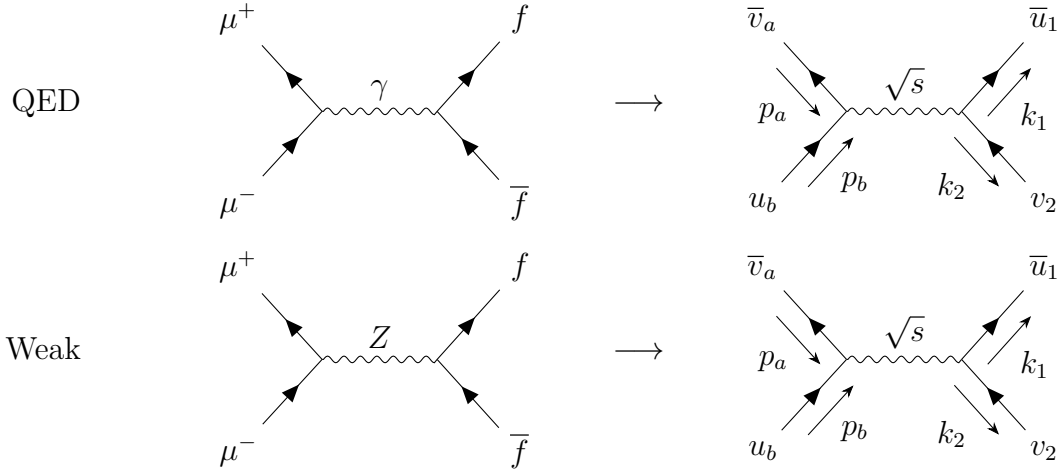
$$\Lambda = 546.71 \quad (4.20)$$

Problem 5 $\mu\bar{\mu} \rightarrow f\bar{f}$

In honor of the recent proposal to build a future muon collider, compute $\mu^+\mu^- \rightarrow f\bar{f}$ (where $f \neq \mu$) at leading order (i.e., tree-level) in the Standard Model.

Assume $m_\mu = m_f = 0$. Determine the numerical value of the production cross-section in fb of $\mu^+\mu^- \rightarrow t\bar{t}$ at $\sqrt{s} = 10$ TeV.

Starting with the Feynman diagrams:



The matrix elements are:

$$\mathcal{M}_{\text{QED}} = [\bar{v}_a(-ie\gamma^\mu)u_b] \frac{-ig_{\mu\nu}}{s} [\bar{u}_1(-ieQ_f\gamma^\nu)v_2] \quad (5.1)$$

$$= \frac{ie^2Q_f}{s} (\bar{v}_a\gamma^\mu u_b)(\bar{u}_1\gamma_\mu v_2) \quad (5.2)$$

$$\mathcal{M}_{\text{QED}}^* = \frac{-ie^2Q_f}{s} (\bar{u}_b\gamma^\mu v_a)(\bar{v}_2\gamma_\mu u_1) \quad (5.3)$$

$$\begin{aligned} \mathcal{M}_{\text{weak}} &= \left[\bar{v}_a \left(-i\gamma^\mu \left(g_V^{(\mu)} - g_A^{(\mu)}\gamma^5 \right) \right) u_b \right] \frac{-i}{s - M_Z^2} \left(g_{\mu\nu} + \frac{p_\nu p_\mu}{M_Z^2} \right) \left[\bar{u}_1 \left(-i\gamma^\nu \left(g_V^f - g_A^f\gamma^5 \right) \right) v_2 \right] \\ &= \frac{i}{s - M_Z^2} \left(g_{\mu\nu} + \frac{p_\nu p_\mu}{M_Z^2} \right) \left[\bar{v}_a \gamma^\mu \left(g_V^{(\mu)} - g_A^{(\mu)}\gamma^5 \right) u_b \right] \left[\bar{u}_1 \gamma^\nu \left(g_V^f - g_A^f\gamma^5 \right) v_2 \right] \end{aligned} \quad (5.4)$$

So now to write $\mathcal{M}_{\mu\bar{\mu} \rightarrow f\bar{f}} = \mathcal{M}_{\text{QED}} + \mathcal{M}_{\text{weak}}$:

$$\begin{aligned} \mathcal{M}_{\mu\bar{\mu} \rightarrow f\bar{f}} &= \frac{ie^2 Q_f}{s} (\bar{v}_a \gamma^\mu u_b) (\bar{u}_1 \gamma_\mu v_2) \\ &+ \frac{i}{s - M_Z^2} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left[\bar{v}_a \gamma^\sigma \left(g_V^{(\mu)} - g_A^{(\mu)} \gamma^5 \right) u_b \right] \left[\bar{u}_1 \gamma^\rho \left(g_V^f - g_A^f \gamma^5 \right) v_2 \right] \end{aligned} \quad (5.5)$$

$$\begin{aligned} \mathcal{M}_{\mu\bar{\mu} \rightarrow f\bar{f}}^* &= \frac{-ie^2 Q_f}{s} (\bar{u}_b \gamma^\mu v_a) (\bar{v}_2 \gamma_\mu u_1) \\ &- \frac{i}{s - M_Z^2} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left[\bar{u}_b \gamma^\sigma \left(g_V^{(\mu)} + g_A^{(\mu)} \gamma^5 \right) v_a \right] \left[\bar{v}_2 \gamma^\rho \left(g_V^f + g_A^f \gamma^5 \right) u_1 \right] \end{aligned} \quad (5.6)$$

Relabelling by index to make the squaring easier:

$$\mathcal{M}_{\mu\bar{\mu} \rightarrow f\bar{f}} = \mathcal{M}_1 + \mathcal{M}_2 \quad (5.7)$$

$$\mathcal{M}_{\mu\bar{\mu} \rightarrow f\bar{f}}^* = \mathcal{M}_1^* + \mathcal{M}_2^* \quad (5.8)$$

$$\frac{1}{4} \sum_{\text{spins}} |\mathcal{M}_{\mu\bar{\mu} \rightarrow f\bar{f}}|^2 = \frac{1}{4} \sum_{\text{spins}} (\mathcal{M}_1 + \mathcal{M}_2)(\mathcal{M}_1^* + \mathcal{M}_2^*) \quad (5.9)$$

$$= \frac{1}{4} \sum_{\text{spins}} \left(\mathcal{M}_1 \mathcal{M}_1^* + \mathcal{M}_1 \mathcal{M}_2^* + \mathcal{M}_1^* \mathcal{M}_2 + \mathcal{M}_2 \mathcal{M}_2^* \right) \quad (5.10)$$

Since the masses are treated to be negligible, the 4-momenta in the CoM frame are:

$$p_a^\mu = \begin{bmatrix} P \\ 0 \\ 0 \\ P \end{bmatrix} \quad p_b^\mu = \begin{bmatrix} P \\ 0 \\ 0 \\ -P \end{bmatrix} \quad k_1^\mu = \begin{bmatrix} K \\ 0 \\ K \sin \theta \\ K \cos \theta \end{bmatrix} \quad k_2^\mu = \begin{bmatrix} K \\ 0 \\ -K \sin \theta \\ -K \cos \theta \end{bmatrix} \quad (5.11)$$

$$P = \sqrt{\frac{s}{4}} \quad K = \sqrt{\frac{s}{4}} \quad (5.12)$$

$$(p_a \cdot p_b) = (k_1 \cdot k_2) = \frac{s}{2} \quad (p_a)^2 = (p_b)^2 = (k_1)^2 = (k_2)^2 = 0 \quad (5.13)$$

$$(p_a \cdot k_1) = \frac{s}{4} (1 - \cos \theta) \quad (p_a \cdot k_2) = \frac{s}{4} (1 + \cos \theta) \quad (5.14)$$

$$(p_b \cdot k_1) = \frac{s}{4} (1 + \cos \theta) \quad (p_b \cdot k_2) = \frac{s}{4} (1 - \cos \theta) \quad (5.15)$$

$$(p_Z \cdot p_a) = (p_a \cdot p_a + p_b \cdot p_a) = \frac{s}{2} \quad (p_Z)^2 = s \quad (5.16)$$

$$\frac{1}{4} \sum_{\text{spins}} \mathcal{M}_1 \mathcal{M}_1^* = \frac{1}{4} \sum_{\text{spins}} \frac{ie^2 Q_f}{s} (\bar{v}_a \gamma^\mu u_b) (\bar{u}_1 \gamma_\mu v_2) \frac{-ie^2 Q_f}{s} (\bar{u}_b \gamma^\nu v_a) (\bar{v}_2 \gamma_\nu u_1) \quad (5.17)$$

$$= \frac{1}{4} \sum_{\text{spins}} \frac{e^4 Q_f^2}{s^2} \text{Tr} [\bar{v}_a \gamma^\mu \not{p}_b \gamma^\nu v_a] \text{Tr} [\bar{u}_1 \gamma_\mu \not{k}_2 \gamma_\nu u_1] \quad (5.18)$$

$$= \frac{e^4 Q_f^2}{4s^2} \text{Tr} [\not{p}_a \gamma^\mu \not{p}_b \gamma^\nu] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma_\nu] \quad (5.19)$$

$$= \frac{e^4 Q_f^2}{4s^2} p_{a,\alpha} p_{b,\beta} k_1^\kappa k_2^\eta 16 (g^{\alpha\mu} g^{\beta\nu} - g^{\alpha\beta} g^{\mu\nu} + g^{\alpha\nu} g^{\mu\beta}) (g_{\kappa\mu} g_{\eta\nu} - g_{\kappa\eta} g_{\mu\nu} + g_{\kappa\nu} g_{\mu\eta}) \quad (5.20)$$

$$= \frac{4e^4 Q_f^2}{s^2} (p_{a,\alpha} p_{b,\beta} g^{\alpha\mu} g^{\beta\nu} - p_{a,\alpha} p_{b,\beta} g^{\alpha\beta} g^{\mu\nu} + p_{a,\alpha} p_{b,\beta} g^{\alpha\nu} g^{\mu\beta}) \dots \quad (5.21)$$

$$\dots (k_1^\kappa k_2^\eta g_{\kappa\mu} g_{\eta\nu} - k_1^\kappa k_2^\eta g_{\kappa\eta} g_{\mu\nu} + k_1^\kappa k_2^\eta g_{\kappa\nu} g_{\mu\eta})$$

$$= \frac{4e^4 Q_f^2}{s^2} (p_a^\mu p_b^\nu - (p_a \cdot p_b) g^{\mu\nu} + p_a^\nu p_b^\mu) (k_{1,\mu} k_{2,\nu} - (k_1 \cdot k_2) g_{\mu\nu} + k_{1,\nu} k_{2,\mu}) \quad (5.22)$$

$$= \frac{4e^4 Q_f^2}{s^2} \left(p_a^\mu p_b^\nu k_{1,\mu} k_{2,\nu} - p_a^\mu p_b^\nu (k_1 \cdot k_2) g_{\mu\nu} + p_a^\mu p_b^\nu k_{1,\nu} k_{2,\mu} \right. \\ \left. - (p_a \cdot p_b) g^{\mu\nu} k_{1,\mu} k_{2,\nu} + (p_a \cdot p_b) g^{\mu\nu} (k_1 \cdot k_2) g_{\mu\nu} - (p_a \cdot p_b) g^{\mu\nu} k_{1,\nu} k_{2,\mu} \right. \\ \left. + p_a^\nu p_b^\mu k_{1,\mu} k_{2,\nu} - p_a^\nu p_b^\mu (k_1 \cdot k_2) g_{\mu\nu} + p_a^\nu p_b^\mu k_{1,\nu} k_{2,\mu} \right) \quad (5.23)$$

$$= \frac{4e^4 Q_f^2}{s^2} \left((p_a \cdot k_1)(p_b \cdot k_2) - (p_a \cdot p_b)(k_1 \cdot k_2) + (p_a \cdot k_2)(p_b \cdot k_1) \right. \\ \left. - (p_a \cdot p_b)(k_1 \cdot k_2) + 4(p_a \cdot p_b)(k_1 \cdot k_2) - (p_a \cdot p_b)(k_1 \cdot k_2) \right. \\ \left. + (p_a \cdot k_2)(p_b \cdot k_1) - (p_a \cdot p_b)(k_1 \cdot k_2) + (p_a \cdot k_1)(p_b \cdot k_2) \right) \quad (5.24)$$

$$= \frac{4e^4 Q_f^2}{s^2} \left(2(p_a \cdot k_1)(p_b \cdot k_2) + 2(p_a \cdot k_2)(p_b \cdot k_1) \right) \quad (5.25)$$

$$= \frac{8e^4 Q_f^2}{s^2} ((p_a \cdot k_1)(p_b \cdot k_2) + (p_a \cdot k_2)(p_b \cdot k_1)) \quad (5.26)$$

$$= \frac{8e^4 Q_f^2}{s^2} \frac{s^2}{16} [(1 - \cos \theta)(1 - \cos \theta) + (1 + \cos \theta)(1 + \cos \theta)] \quad (5.27)$$

$$\boxed{\frac{1}{4} \sum_{\text{spins}} \mathcal{M}_1 \mathcal{M}_1^* = e^4 Q_f^2 (1 + \cos^2 \theta)} \quad (5.28)$$

$$\frac{1}{4} \sum_{\text{spins}} \mathcal{M}_1 \mathcal{M}_2^* = \frac{1}{4} \sum_{\text{spins}} \frac{ie^2 Q_f}{s} (\bar{v}_a \gamma^\mu u_b) (\bar{u}_1 \gamma_\mu v_2) \frac{-i}{s - M_Z^2} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \times \dots \quad (5.29)$$

$$\times \left[\bar{u}_b \gamma^\sigma \left(g_V^{(\mu)} + g_A^{(\mu)} \gamma^5 \right) v_a \right] \left[\bar{v}_2 \gamma^\rho \left(g_V^f + g_A^f \gamma^5 \right) u_1 \right] \\ = \frac{1}{4} \sum_{\text{spins}} \frac{e^2 Q_f}{s(s - M_Z^2)} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) (\bar{v}_a \gamma^\mu u_b) (\bar{u}_1 \gamma_\mu v_2) \times \dots \quad (5.30)$$

$$\times \left[\bar{u}_b \gamma^\sigma \left(g_V^{(\mu)} + g_A^{(\mu)} \gamma^5 \right) v_a \right] \left[\bar{v}_2 \gamma^\rho \left(g_V^f + g_A^f \gamma^5 \right) u_1 \right] \\ = \frac{e^2 Q_f}{4s(s - M_Z^2)} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \left(g_V^{(\mu)} + g_A^{(\mu)} \gamma^5 \right) \right] \times \dots \quad (5.31)$$

$$\times \text{Tr} \left[\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho \left(g_V^f + g_A^f \gamma^5 \right) \right] \\ = \boxed{\frac{e^2 Q_f}{4s(s - M_Z^2)}} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left(g_V^{(\mu)} \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \right] + g_A^{(\mu)} \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \gamma^5 \right] \right) \\ \times \left(g_V^f \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho] + g_A^f \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho \gamma^5] \right) \quad (5.32)$$

$$= \boxed{A_f} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left(g_V^{(\mu)} \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \right] + g_A^{(\mu)} \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \gamma^5 \right] \right) \times \dots \quad (5.33)$$

$$\times \left(g_V^f \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho] + g_A^f \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho \gamma^5] \right) \\ = A_f \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left(g_V^{(\mu)} g_V^f \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \right] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho] + \dots \right. \quad (5.34)$$

$$+ g_A^{(\mu)} g_A^f \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \gamma^5 \right] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho \gamma^5] \Big) \\ = A_f \left\{ g_V^{(\mu)} g_V^f \left(g_{\sigma\rho} \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \right] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho] + \dots \right. \right. \\ + \frac{p_\rho p_\sigma}{M_Z^2} \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \right] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho] \Big) + \dots \\ + g_A^{(\mu)} g_A^f \left(g_{\sigma\rho} \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \gamma^5 \right] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho \gamma^5] + \dots \right. \\ + \frac{p_\rho p_\sigma}{M_Z^2} \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \gamma^5 \right] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho \gamma^5] \Big) \Big\} \quad (5.35)$$

$$= A_f \left\{ g_V^{(\mu)} g_V^f \left(\text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \right] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\sigma] + \dots \right. \right. \\ + \frac{1}{M_Z^2} \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \not{p}_Z \right] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \not{p}_Z] \Big) + \dots \\ + g_A^{(\mu)} g_A^f \left(\text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \gamma^5 \right] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\sigma \gamma^5] + \dots \right. \\ + \frac{1}{M_Z^2} \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \not{p}_Z \gamma^5 \right] \text{Tr} [\not{k}_1 \gamma_\mu \not{k}_2 \not{p}_Z \gamma^5] \Big) \Big\} \quad (5.36)$$

Checking the traces:

$$\text{Tr}[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma] \text{Tr}[\not{k}_1 \gamma_\mu \not{k}_2 \gamma_\sigma] = \boxed{4s^2(1 + \cos^2 \theta)} \quad (5.37)$$

$$\text{Tr}[\not{p}_a \gamma^\mu \not{p}_b \not{p}_Z] \text{Tr}[\not{k}_1 \gamma_\mu \not{k}_2 \not{p}_Z] = p_{a,\alpha} p_{b,\beta} p_{Z,\zeta} k_1^\kappa k_2^\eta p_Z^\lambda \text{Tr}[\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\zeta] \text{Tr}[\gamma_\kappa \gamma_\mu \gamma_\eta \gamma_\lambda] \quad (5.38)$$

$$\begin{aligned} &= 16 p_{a,\alpha} p_{b,\beta} p_{Z,\zeta} k_1^\kappa k_2^\eta p_Z^\lambda \times \dots \\ &\dots \times (g^{\alpha\mu} g^{\beta\zeta} - g^{\alpha\beta} g^{\mu\zeta} + g^{\alpha\zeta} g^{\mu\beta}) (g_{\kappa\mu} g_{\eta\lambda} - g_{\kappa\eta} g_{\mu\lambda} + g_{\kappa\lambda} g_{\mu\eta}) \end{aligned} \quad (5.39)$$

$$\begin{aligned} &= 16 p_{a,\alpha} p_{b,\beta} p_{Z,\zeta} (g^{\alpha\mu} g^{\beta\zeta} g^{\alpha\beta} g^{\mu\zeta} g^{\alpha\zeta} g^{\mu\beta}) \\ &\dots \times (k_1^\kappa k_2^\eta p_Z^\lambda g_{\kappa\mu} g_{\eta\lambda} - k_1^\kappa k_2^\eta p_Z^\lambda g_{\kappa\eta} g_{\mu\lambda} + k_1^\kappa k_2^\eta p_Z^\lambda g_{\kappa\lambda} g_{\mu\eta}) \end{aligned} \quad (5.40)$$

$$\begin{aligned} &= 16 (p_{a,\alpha} (p_Z \cdot p_b) g^{\alpha\mu} - (p_a \cdot p_b) p_{Z,\mu} + p_b^\mu (p_Z \cdot p_a)) \times \dots \\ &\dots \times (k_1^\kappa (p_Z \cdot k_2) g_{\kappa\mu} - (k_1 \cdot k_2) p_{Z,\mu} + (p_Z \cdot k_1) k_{2,\mu}) \end{aligned} \quad (5.41)$$

$$= 16 \left(p_Z^\mu \frac{s}{2} - \frac{s}{2} p_Z^\mu \right) \left(p_{Z,\mu} \frac{s}{2} - \frac{s}{2} p_{Z,\mu} \right) \quad (5.42)$$

$$\boxed{0} \quad (5.43)$$

Checking the other two traces:

$$\text{Tr}[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \gamma^5] \text{Tr}[\not{k}_1 \gamma_\mu \not{k}_2 \gamma_\sigma \rho \gamma^5] = p_{a,\alpha} p_{b,\beta} k_1^\kappa k_2^\eta \text{Tr}[\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\sigma \gamma^5] \text{Tr}[\gamma_\kappa \gamma_\mu \gamma_\eta \gamma_\sigma \gamma^5] \quad (5.44)$$

$$= p_{a,\alpha} p_{b,\beta} k_1^\kappa k_2^\eta (-16) \epsilon^{\alpha\mu\beta\sigma} \epsilon_{\kappa\mu\eta\sigma} \quad (5.45)$$

$$= p_{a,\alpha} p_{b,\beta} k_1^\kappa k_2^\eta (-16) (-2\delta_\kappa^\alpha \delta_\eta^\beta + 2\delta_\eta^\alpha \delta_\kappa^\beta) \quad (5.46)$$

$$= 32 ((p_a \cdot k_1)(p_b \cdot k_2) - (p_a \cdot k_2)(p_b \cdot k_1)) \quad (5.47)$$

$$\begin{aligned} &= 32 \left(\frac{s}{4} (1 - \cos \theta) \frac{s}{4} (1 - \cos \theta) - \dots \right. \\ &\quad \left. \dots - \frac{s}{4} (1 + \cos \theta) \frac{s}{4} (1 + \cos \theta) \right) \end{aligned} \quad (5.48)$$

$$= 32 \left(\frac{s^2}{16} (1 - 2 \cos \theta + \cos^2 \theta) - \frac{s^2}{16} (1 + 2 \cos \theta + \cos^2 \theta) \right) \quad (5.49)$$

$$= 2s^2 \left((1 - 2 \cos \theta + \cos^2 \theta) - \frac{s^2}{4} (1 + 2 \cos \theta + \cos^2 \theta) \right) \quad (5.50)$$

$$= \boxed{-4s^2 \cos \theta} \quad (5.51)$$

$$\text{Tr}[\not{p}_a \gamma^\mu \not{p}_b \not{p}_Z \gamma^5] \text{Tr}[\not{k}_1 \gamma_\mu \not{k}_2 \not{p}_Z \gamma^5] = p_{a,\alpha} p_{b,\beta} p_{Z,\zeta} k_1^\kappa k_2^\eta p_Z^\lambda \text{Tr}[\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\zeta \gamma^5] \text{Tr}[\gamma_\kappa \gamma_\mu \gamma_\eta \gamma_\lambda \gamma^5] \quad (5.52)$$

$$= p_{a,\alpha} p_{b,\beta} p_{Z,\zeta} k_1^\kappa k_2^\eta p_Z^\lambda (-16) \epsilon^{\alpha\mu\beta\zeta} \epsilon_{\kappa\mu\eta\lambda} \quad (5.53)$$

$$= \boxed{\text{F.F.}} \quad (5.54)$$

Plugging these results back in to $\frac{1}{4} \sum_{\text{spins}} \mathcal{M}_1 \mathcal{M}_2^*$

$$\begin{aligned} \frac{1}{4} \sum_{\text{spins}} \mathcal{M}_1 \mathcal{M}_2^* &= A_f \left\{ g_V^{(\mu)} g_V^f (4s^2 (1 + \cos^2 \theta) + 0) + \dots \right. \\ &\quad \left. + g_A^{(\mu)} g_A^f \left(-4s^2 \cos \theta + \frac{\text{F.F.}}{M_Z^2} \right) \right\} \end{aligned} \quad (5.55)$$

$$\begin{aligned} &= \frac{e^2 Q_f}{4s(s - M_Z^2)} \left\{ 4s^2 \left(g_V^{(\mu)} g_V^f - g_A^{(\mu)} g_A^f \cos \theta + g_V^{(\mu)} g_V^f \cos^2 \theta \right) + \dots \right. \\ &\quad \left. + g_A^{(\mu)} g_A^f \frac{\text{F.F.}}{M_Z^2} \right\} \end{aligned} \quad (5.56)$$

$$\begin{aligned} \frac{1}{4} \sum_{\text{spins}} \mathcal{M}_1 \mathcal{M}_2^* &= \frac{e^2 Q_f}{4s(s - M_Z^2)} \left\{ 4s^2 \left(g_V^{(\mu)} g_V^f - g_A^{(\mu)} g_A^f \cos \theta + g_V^{(\mu)} g_V^f \cos^2 \theta \right) + \dots \right. \\ &\quad \left. \dots + g_A^{(\mu)} g_A^f \frac{\text{F.F.}}{M_Z^2} \right\} \end{aligned}$$

(5.57)

$$\frac{1}{4} \sum_{\text{spins}} \mathcal{M}_2 \mathcal{M}_1^* = \frac{1}{4} \sum_{\text{spins}} \frac{eQ_f}{s(s - M_Z^2)} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left[\bar{v}_a \gamma^\sigma \left(g_V^{(\mu)} - g_A^{(\mu)} \gamma^5 \right) u_b \right] \times \dots \quad (5.58)$$

$$\dots \times \left[\bar{u}_1 \gamma^\rho \left(g_V^f - g_A^f \gamma^5 \right) v_2 \right] (\bar{u}_b \gamma^\mu v_a) (\bar{v}_2 \gamma_\mu u_1) \\ = \boxed{\frac{eQ_f}{4s(s - M_Z^2)}} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left[\text{Tr} \left[\not{p}_a \gamma^\sigma \left(g_V^{(\mu)} - g_A^{(\mu)} \gamma^5 \right) \not{p}_b \gamma^\mu \right] \times \dots \right. \quad (5.59)$$

$$\dots \times \text{Tr} \left[\not{k}_1 \gamma^\rho \left(g_V^f - g_A^f \gamma^5 \right) \not{k}_2 \gamma_\mu \right] \Big] \\ = \boxed{A_f} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left[\text{Tr} \left[\not{p}_a \gamma^\sigma \not{p}_b \gamma^\mu \left(g_V^{(\mu)} - g_A^{(\mu)} \gamma^5 \right) \right] \times \dots \right. \quad (5.60)$$

$$\dots \times \text{Tr} \left[\not{k}_1 \gamma^\rho \not{k}_2 \gamma_\mu \left(g_V^f - g_A^f \gamma^5 \right) \right] \Big] \\ = \frac{eQ_f}{4s(s - M_Z^2)} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left[\left(g_V^{(\mu)} \text{Tr} \left[\not{p}_a \gamma^\sigma \not{p}_b \gamma^\mu \right] - g_A^{(\mu)} \text{Tr} \left[\not{p}_a \gamma^\sigma \not{p}_b \gamma^\mu \gamma^5 \right] \right) \times \dots \right. \quad (5.61)$$

$$\dots \times \left(g_V^f \text{Tr} \left[\not{k}_1 \gamma^\rho \not{k}_2 \gamma_\mu \right] - g_A^f \text{Tr} \left[\not{k}_1 \gamma^\rho \not{k}_2 \gamma_\mu \gamma^5 \right] \right) \Big] \\ = \frac{eQ_f}{4s(s - M_Z^2)} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left[g_V^{(\mu)} g_V^f \text{Tr} \left[\not{p}_a \gamma^\sigma \not{p}_b \gamma^\mu \right] \text{Tr} \left[\not{k}_1 \gamma^\rho \not{k}_2 \gamma_\mu \right] + \dots \right. \quad (5.62)$$

$$\dots + g_A^{(\mu)} g_A^f \text{Tr} \left[\not{p}_a \gamma^\sigma \not{p}_b \gamma^\mu \gamma^5 \right] \text{Tr} \left[\not{k}_1 \gamma^\rho \not{k}_2 \gamma_\mu \gamma^5 \right] \Big]$$

Hold on, that looks familiar.. almost like Eqn 5.34:

$$= A_f \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left(g_V^{(\mu)} g_V^f \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \right] \text{Tr} \left[\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho \right] + \dots \right. \quad (5.34)$$

$$\left. + g_A^{(\mu)} g_A^f \text{Tr} \left[\not{p}_a \gamma^\mu \not{p}_b \gamma^\sigma \gamma^5 \right] \text{Tr} \left[\not{k}_1 \gamma_\mu \not{k}_2 \gamma^\rho \gamma^5 \right] \right)$$

They aren't exactly the same as I've

$$\begin{aligned}
\frac{1}{4} \sum_{\text{spins}} \mathcal{M}_2 \mathcal{M}_2^* &= \sum_{\text{spins}} \frac{1}{4(s - M_Z^2)^2} \\
&\dots \times \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left[\bar{v}_a \gamma^\sigma \left(g_V^{(\mu)} - g_A^{(\mu)} \gamma^5 \right) u_b \right] \left[\bar{u}_1 \gamma^\rho \left(g_V^f - g_A^f \gamma^5 \right) v_2 \right] \quad (5.63) \\
&\dots \times \left(g_{\lambda\delta} + \frac{p_\delta p_\lambda}{M_Z^2} \right) \left[\bar{u}_b \gamma^\lambda \left(g_V^{(\mu)} + g_A^{(\mu)} \gamma^5 \right) v_a \right] \left[\bar{v}_2 \gamma^\delta \left(g_V^f + g_A^f \gamma^5 \right) u_1 \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4(s - M_Z^2)^2} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left(g_{\lambda\delta} + \frac{p_\delta p_\lambda}{M_Z^2} \right) \\
&\dots \times \text{Tr} \left[\not{p}_a \gamma^\sigma \left(g_V^{(\mu)} - g_A^{(\mu)} \gamma^5 \right) \not{p}_b \gamma^\lambda \left(g_V^{(\mu)} + g_A^{(\mu)} \gamma^5 \right) \right] \text{Tr} \left[\not{k}_1 \gamma^\rho \left(g_V^f - g_A^f \gamma^5 \right) \not{k}_2 \gamma^\delta \left(g_V^f + g_A^f \gamma^5 \right) \right] \quad (5.64)
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4(s - M_Z^2)^2} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left(g_{\lambda\delta} + \frac{p_\delta p_\lambda}{M_Z^2} \right) \\
&\dots \times \text{Tr} \left[\not{p}_a \gamma^\sigma \left(g_V^{(\mu)} - g_A^{(\mu)} \gamma^5 \right) \not{p}_b \gamma^\lambda \left(g_V^{(\mu)} + g_A^{(\mu)} \gamma^5 \right) \right] \text{Tr} \left[\not{k}_1 \gamma^\rho \left(g_V^f - g_A^f \gamma^5 \right) \not{k}_2 \gamma^\delta \left(g_V^f + g_A^f \gamma^5 \right) \right] \quad (5.65)
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4(s - M_Z^2)^2} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left(g_{\lambda\delta} + \frac{p_\delta p_\lambda}{M_Z^2} \right) \\
&\dots \times \text{Tr} \left[\not{p}_a \gamma^\sigma \not{p}_b \gamma^\lambda \left(g_V^{(\mu)} - g_A^{(\mu)} \gamma^5 \right) \left(g_V^{(\mu)} + g_A^{(\mu)} \gamma^5 \right) \right] \text{Tr} \left[\not{k}_1 \gamma^\rho \not{k}_2 \gamma^\delta \left(g_V^f - g_A^f \gamma^5 \right) \left(g_V^f + g_A^f \gamma^5 \right) \right] \quad (5.66)
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4(s - M_Z^2)^2} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \left(g_{\lambda\delta} + \frac{p_\delta p_\lambda}{M_Z^2} \right) \text{Tr} \left[\not{p}_a \gamma^\sigma \not{p}_b \gamma^\lambda \left(g_V^{(\mu)} g_V^{(\mu)} - g_A^{(\mu)} g_A^{(\mu)} \right) \right] \quad (5.67) \\
&\dots \times \text{Tr} \left[\not{k}_1 \gamma^\rho \not{k}_2 \gamma^\delta \left(g_V^f g_V^f - g_A^f g_A^f \right) \right]
\end{aligned}$$

$$\begin{aligned}
&= \boxed{\frac{1}{4(s - M_Z^2)^2} \left(g_V^{(\mu)} g_V^{(\mu)} - g_A^{(\mu)} g_A^{(\mu)} \right) \left(g_V^f g_V^f - g_A^f g_A^f \right)} \left(g_{\sigma\rho} + \frac{p_\rho p_\sigma}{M_Z^2} \right) \text{Tr} \left[\not{p}_a \gamma^\sigma \not{p}_b \gamma^\lambda \right] \quad (5.68) \\
&\dots \times \left(g_{\lambda\delta} + \frac{p_\delta p_\lambda}{M_Z^2} \right) \text{Tr} \left[\not{k}_1 \gamma^\rho \not{k}_2 \gamma^\delta \right]
\end{aligned}$$

$$\begin{aligned}
&= \boxed{C_f} \left(g_{\sigma\rho} g_{\lambda\delta} + g_{\sigma\rho} \frac{p_{Z,\delta} p_{Z,\lambda}}{M_Z^2} + \frac{p_{Z,\rho} p_{Z,\sigma}}{M_Z^2} g_{\lambda\delta} + \frac{p_{Z,\rho} p_{Z,\sigma} p_{Z,\delta} p_{Z,\lambda}}{M_Z^4} \right) \text{Tr} \left[\not{p}_a \gamma^\sigma \not{p}_b \gamma^\lambda \right] \text{Tr} \left[\not{k}_1 \gamma^\rho \not{k}_2 \gamma^\delta \right] \quad (5.69)
\end{aligned}$$

$$\begin{aligned}
&= C_f \left(g_{\sigma\rho} g_{\lambda\delta} + g_{\sigma\rho} \frac{p_{Z,\delta} p_{Z,\lambda}}{M_Z^2} + \frac{p_{Z,\rho} p_{Z,\sigma}}{M_Z^2} g_{\lambda\delta} + \frac{p_{Z,\rho} p_{Z,\sigma} p_{Z,\delta} p_{Z,\lambda}}{M_Z^4} \right) p_{a,\alpha} p_{b,\beta} k_{1,\kappa} k_{2,\eta} \times \dots \quad (5.70) \\
&\dots \times \text{Tr} \left[\gamma^\alpha \gamma^\sigma \gamma^\beta \gamma^\lambda \right] \text{Tr} \left[\gamma^\kappa \gamma^\rho \gamma^\eta \gamma^\delta \right]
\end{aligned}$$

$$\begin{aligned}
&= C_f \left(g_{\sigma\rho} g_{\lambda\delta} + g_{\sigma\rho} \frac{p_{Z,\delta} p_{Z,\lambda}}{M_Z^2} + \frac{p_{Z,\rho} p_{Z,\sigma}}{M_Z^2} g_{\lambda\delta} + \frac{p_{Z,\rho} p_{Z,\sigma} p_{Z,\delta} p_{Z,\lambda}}{M_Z^4} \right) p_{a,\alpha} p_{b,\beta} k_{1,\kappa} k_{2,\eta} \times \dots \quad (5.71) \\
&\dots \times 16 (g^{\alpha\sigma} g^{\beta\lambda} - g^{\alpha\beta} g^{\sigma\lambda} + g^{\alpha\lambda} g^{\sigma\beta}) (g^{\kappa\rho} g^{\eta\delta} - g^{\kappa\eta} g^{\rho\delta} + g^{\kappa\delta} g^{\rho\eta})
\end{aligned}$$

$$\frac{1}{4} \sum_{\text{spins}} \mathcal{M}_2 \mathcal{M}_2^* = 16C_f \left(g_{\sigma\rho} g_{\lambda\delta} + g_{\sigma\rho} \frac{p_{Z,\delta} p_{Z,\lambda}}{M_Z^2} + \frac{p_{Z,\rho} p_{Z,\sigma}}{M_Z^2} g_{\lambda\delta} + \frac{p_{Z,\rho} p_{Z,\sigma} p_{Z,\delta} p_{Z,\lambda}}{M_Z^4} \right) \times \dots$$

$$\times p_{a,\alpha} p_{b,\beta} k_{1,\kappa} k_{2,\eta} (g^{\alpha\sigma} g^{\beta\lambda} - g^{\alpha\beta} g^{\sigma\lambda} + g^{\alpha\lambda} g^{\sigma\beta}) (g^{\kappa\rho} g^{\eta\delta} - g^{\kappa\eta} g^{\rho\delta} + g^{\kappa\delta} g^{\rho\eta})$$
(5.72)

$$= 16C_f \left(g_{\sigma\rho} g_{\lambda\delta} + g_{\sigma\rho} \frac{p_{Z,\delta} p_{Z,\lambda}}{M_Z^2} + \frac{p_{Z,\rho} p_{Z,\sigma}}{M_Z^2} g_{\lambda\delta} + \frac{p_{Z,\rho} p_{Z,\sigma} p_{Z,\delta} p_{Z,\lambda}}{M_Z^4} \right) \times \dots$$

$$\dots \times (p_{a,\alpha} p_{b,\beta} g^{\alpha\sigma} g^{\beta\lambda} - p_{a,\alpha} p_{b,\beta} g^{\alpha\beta} g^{\sigma\lambda} + p_{a,\alpha} p_{b,\beta} g^{\alpha\lambda} g^{\sigma\beta}) \times \dots$$

$$\dots \times (k_{1,\kappa} k_{2,\eta} g^{\kappa\rho} g^{\eta\delta} - k_{1,\kappa} k_{2,\eta} g^{\kappa\eta} g^{\rho\delta} + k_{1,\kappa} k_{2,\eta} g^{\kappa\delta} g^{\rho\eta})$$
(5.73)

$$= 16C_f \left(g_{\sigma\rho} g_{\lambda\delta} + g_{\sigma\rho} \frac{p_{Z,\delta} p_{Z,\lambda}}{M_Z^2} + \frac{p_{Z,\rho} p_{Z,\sigma}}{M_Z^2} g_{\lambda\delta} + \frac{p_{Z,\rho} p_{Z,\sigma} p_{Z,\delta} p_{Z,\lambda}}{M_Z^4} \right) \times \dots$$

$$\dots \times (p_a^\sigma p_b^\lambda - (p_a \cdot p_b) g^{\sigma\lambda} + p_a^\lambda p_b^\sigma) (k_1^\rho k_2^\delta - (k_1 \cdot k_2) g^{\rho\delta} + k_1^\delta k_2^\rho)$$
(5.74)

$$= 16C_f \left(g_{\sigma\rho} g_{\lambda\delta} + g_{\sigma\rho} \frac{p_{Z,\delta} p_{Z,\lambda}}{M_Z^2} + \frac{p_{Z,\rho} p_{Z,\sigma}}{M_Z^2} g_{\lambda\delta} + \frac{p_{Z,\rho} p_{Z,\sigma} p_{Z,\delta} p_{Z,\lambda}}{M_Z^4} \right) \times \dots$$

$$\dots \times \left(p_a^\sigma p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} + p_a^\lambda p_b^\sigma \right) \left(k_1^\rho k_2^\delta - \frac{s}{2} g^{\rho\delta} + k_1^\delta k_2^\rho \right)$$
(5.75)

$$= 16C_f \left[g_{\sigma\rho} g_{\lambda\delta} \left(p_a^\sigma p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} + p_a^\lambda p_b^\sigma \right) + \dots \right.$$

$$\dots + g_{\sigma\rho} \frac{p_{Z,\delta} p_{Z,\lambda}}{M_Z^2} \left(p_a^\sigma p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} + p_a^\lambda p_b^\sigma \right) + \dots$$

$$\dots + \frac{p_{Z,\rho} p_{Z,\sigma}}{M_Z^2} g_{\lambda\delta} \left(p_a^\sigma p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} + p_a^\lambda p_b^\sigma \right) + \dots$$

$$\left. \dots + \frac{p_{Z,\rho} p_{Z,\sigma} p_{Z,\delta} p_{Z,\lambda}}{M_Z^4} \left(p_a^\sigma p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} + p_a^\lambda p_b^\sigma \right) \right] \times \dots$$

$$\dots \times \left(k_1^\rho k_2^\delta - \frac{s}{2} g^{\rho\delta} + k_1^\delta k_2^\rho \right)$$
(5.76)

$$= 16C_f \left[g_{\lambda\delta} \left(p_{a,\rho} p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} g_{\sigma\rho} + p_a^\lambda p_{b,\rho} \right) + \dots \right.$$

$$\dots + g_{\sigma\rho} \frac{p_{Z,\delta}}{M_Z^2} \left(p_a^\sigma \frac{s}{2} - \frac{s}{2} p_{Z,\sigma} + p_b^\sigma \frac{s}{2} \right) + \dots$$

$$\left. \dots + \frac{p_{Z,\rho}}{M_Z^2} g_{\lambda\delta} \left(p_b^\lambda \frac{s}{2} - \frac{s}{2} g^{\sigma\lambda} p_{Z,\sigma} + p_a^\lambda \frac{s}{2} \right) + \frac{p_{Z,\rho} p_{Z,\delta}}{M_Z^4} \left(\frac{s^2}{2} - \frac{s}{2} (p_Z \cdot p_Z) \right) \right] \times \dots$$

$$\dots \times \left(k_1^\rho k_2^\delta - \frac{s}{2} g^{\rho\delta} + k_1^\delta k_2^\rho \right)$$
(5.77) ■

$$\begin{aligned}
&= 16C_f \left[g_{\lambda\delta} \left(p_{a,\rho} p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} g_{\sigma\rho} + p_a^\lambda p_{b,\rho} \right) + g_{\sigma\rho} \frac{p_{Z,\delta}}{M_Z^2} \left(p_Z^\sigma \frac{s}{2} - \frac{s}{2} p_Z^\sigma \right) + \dots \right. \\
&\quad \left. \dots + \frac{p_{Z,\rho}}{M_Z^2} g_{\lambda\delta} \left(p_Z^\lambda \frac{s}{2} - \frac{s}{2} g^{\sigma\lambda} p_{Z,\sigma} \right) + \frac{p_{Z,\rho} p_{Z,\delta}}{M_Z^4} \left(\frac{s^2}{2} - \frac{s}{2} (p_Z \cdot p_Z) \right) \right] \times \dots \\
&\quad \dots \times \left(k_1^\rho k_2^\delta - \frac{s}{2} g^{\rho\delta} + k_1^\delta k_2^\rho \right)
\end{aligned} \tag{5.78}$$

$$\begin{aligned}
&= 16C_f \left[g_{\lambda\delta} \left(p_{a,\rho} p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} g_{\sigma\rho} + p_a^\lambda p_{b,\rho} \right) + 0 + \dots \right. \\
&\quad \left. \dots + \frac{p_{Z,\rho}}{M_Z^2} g_{\lambda\delta} \left(p_Z^\lambda \frac{s}{2} - \frac{s}{2} g^{\sigma\lambda} p_{Z,\sigma} \right) + \frac{p_{Z,\rho} p_{Z,\delta}}{M_Z^4} \left(\frac{s^2}{2} - \frac{s^2}{2} \right) \right] \times \dots \\
&\quad \dots \times \left(k_1^\rho k_2^\delta - \frac{s}{2} g^{\rho\delta} + k_1^\delta k_2^\rho \right)
\end{aligned} \tag{5.79}$$

$$\begin{aligned}
&= 16C_f \left[g_{\lambda\delta} \left(p_{a,\rho} p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} g_{\sigma\rho} + p_a^\lambda p_{b,\rho} \right) + \frac{p_{Z,\rho}}{M_Z^2} g_{\lambda\delta} \left(p_Z^\lambda \frac{s}{2} - \frac{s}{2} g^{\sigma\lambda} p_{Z,\sigma} \right) + 0 \right] \times \dots \\
&\quad \dots \times \left(k_1^\rho k_2^\delta - \frac{s}{2} g^{\rho\delta} + k_1^\delta k_2^\rho \right)
\end{aligned} \tag{5.80}$$

$$\begin{aligned}
&= 16C_f \left[k_1^\rho k_2^\delta g_{\lambda\delta} \left(p_{a,\rho} p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} g_{\sigma\rho} + p_a^\lambda p_{b,\rho} \right) + k_1^\rho k_2^\delta \frac{p_{Z,\rho}}{M_Z^2} g_{\lambda\delta} \left(p_Z^\lambda \frac{s}{2} - \frac{s}{2} g^{\sigma\lambda} p_{Z,\sigma} \right) + \dots \right. \\
&\quad \dots - \frac{s}{2} g^{\rho\delta} g_{\lambda\delta} \left(p_{a,\rho} p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} g_{\sigma\rho} + p_a^\lambda p_{b,\rho} \right) - \frac{s}{2} g^{\rho\delta} \frac{p_{Z,\rho}}{M_Z^2} g_{\lambda\delta} \left(p_Z^\lambda \frac{s}{2} - \frac{s}{2} g^{\sigma\lambda} p_{Z,\sigma} \right) + \dots \\
&\quad \left. \dots + k_1^\delta k_2^\rho g_{\lambda\delta} \left(p_{a,\rho} p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} g_{\sigma\rho} + p_a^\lambda p_{b,\rho} \right) + k_1^\delta k_2^\rho \frac{p_{Z,\rho}}{M_Z^2} g_{\lambda\delta} \left(p_Z^\lambda \frac{s}{2} - \frac{s}{2} g^{\sigma\lambda} p_{Z,\sigma} \right) \right]
\end{aligned} \tag{5.81}$$

$$\begin{aligned}
&= 16C_f \left[k_1^\rho k_{2,\lambda} \left(p_{a,\rho} p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} g_{\sigma\rho} + p_a^\lambda p_{b,\rho} \right) + k_1^\rho k_2^\delta \frac{p_{Z,\rho}}{M_Z^2} g_{\lambda\delta} \left(p_Z^\lambda \frac{s}{2} - \frac{s}{2} g^{\sigma\lambda} p_{Z,\sigma} \right) + \dots \right. \\
&\quad \dots - \frac{s}{2} g^{\rho\delta} g_{\lambda\delta} \left(p_{a,\rho} p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} g_{\sigma\rho} + p_a^\lambda p_{b,\rho} \right) - \frac{s}{2} g^{\rho\delta} \frac{p_{Z,\rho}}{M_Z^2} g_{\lambda\delta} \left(p_Z^\lambda \frac{s}{2} - \frac{s}{2} g^{\sigma\lambda} p_{Z,\sigma} \right) + \dots \\
&\quad \left. \dots + k_{1,\lambda} k_2^\rho \left(p_{a,\rho} p_b^\lambda - \frac{s}{2} g^{\sigma\lambda} g_{\sigma\rho} + p_a^\lambda p_{b,\rho} \right) + k_{1,\lambda} \frac{(p_Z \cdot k_2)}{M_Z^2} \left(p_Z^\lambda \frac{s}{2} - \frac{s}{2} g^{\sigma\lambda} p_{Z,\sigma} \right) \right]
\end{aligned} \tag{5.82}$$

$$\begin{aligned}
&= 16C_f \left[\left((p_a \cdot k_1)(p_b \cdot k_2) - \frac{s}{2}(k_1 \cdot k_2) + (p_a \cdot k_2)(p_b \cdot k_1) \right) + \dots \right. \\
&\quad \dots + \frac{(p_Z \cdot k_1)}{M_Z^2} \left((p_Z \cdot k_2) \frac{s}{2} - \frac{s}{2}(p_Z \cdot k_2) \right) + \dots \\
&\quad \dots + \left(-(p_a \cdot p_b) + 4\frac{s}{2} - (p_a \cdot p_b) \right) - \frac{s}{2} \frac{1}{M_Z^2} \left((p_Z)^2 \frac{s}{2} - \frac{s}{2}(p_Z)^2 \right) + \dots \\
&\quad \dots \left((p_a \cdot k_2)(p_b \cdot k_1) - \frac{s}{2}(k_1 \cdot k_2) + (p_a \cdot k_1)(p_b \cdot k_2) \right) + \dots \\
&\quad \left. \dots + \frac{(p_Z \cdot k_2)}{M_Z^2} \left((p_Z \cdot k_1) \frac{s}{2} - \frac{s}{2}(p_Z \cdot k_1) \right) \right] \quad (5.83)
\end{aligned}$$

$$\begin{aligned}
&= 16C_f \left[2 \left((p_a \cdot k_1)(p_b \cdot k_2) - \frac{s^2}{4} + (p_a \cdot k_2)(p_b \cdot k_1) \right) + \dots \right. \\
&\quad \left. \dots + \left(-(p_a \cdot p_b) + 4\frac{s}{2} - (p_a \cdot p_b) \right) \right] \quad (5.84)
\end{aligned}$$

$$\begin{aligned}
&= 16C_f \left[2 \left(\frac{s^2}{4} (1 + \cos^2 \theta) - \frac{s^2}{4} + \frac{s^2}{4} (1 + \cos^2 \theta) \right) + \dots \right. \\
&\quad \left. \dots + \left(-\frac{s}{2} + 4\frac{s}{2} - \frac{s}{2} \right) \right] \quad (5.85)
\end{aligned}$$

$$= 16C_f \left[\frac{s^2}{2} + s^2 \cos^2 \theta + s \right] \quad (5.86)$$

$$= 16 \frac{1}{4(s - M_Z^2)^2} \left(g_V^{(\mu)} g_V^{(\mu)} - g_A^{(\mu)} g_A^{(\mu)} \right) \left(g_V^f g_V^f - g_A^f g_A^f \right) \left[\frac{s^2}{2} + s^2 \cos^2 \theta + s \right] \quad (5.87)$$

$$\boxed{\frac{1}{4} \sum_{\text{spins}} \mathcal{M}_2 \mathcal{M}_2^* = \frac{4}{(s - M_Z^2)^2} \left(g_V^{(\mu)} g_V^{(\mu)} - g_A^{(\mu)} g_A^{(\mu)} \right) \left(g_V^f g_V^f - g_A^f g_A^f \right) \left[s^2 \left(\frac{1}{2} + \cos^2 \theta \right) + s \right]} \quad (5.88)$$